



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B. TECH
CIVIL ENGINEERING

3rd SEMESTER



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Course Structure (From Academic Session 2024-25 onwards)

B. Tech 3rd Semester: Civil Engineering

Semester III

Sl. No.	Course Code	Course	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	CE241301	Fluid Mechanics	3	0	0	3	30	70
2	CE241302	Engineering Survey	3	0	0	3	30	70
3	CE241303	Construction Materials and Concrete Technology	3	0	0	3	30	70
4	CE241304	Solid Mechanics	3	0	0	3	30	70
5	MA241305A	Mathematics III A	3	0	0	3	30	70
6	HS241306	Indian Knowledge Systems	2	0	0	2	-	50
7	AC241307	Language other than MIL	1	1	0	0	-	50
Practical								
1	CE241317	BCP Laboratory	0	1	2	2	15	35
2	CE241313	Construction Materials and Concrete Technology Lab	0	0	2	1	15	35
3	CE241312	Engineering Survey Lab	0	0	2	1	15	35
4	SI241321	Social Internship	0	0	4	2	-	50
TOTAL			18	2	10	23	195	605
Total Contact Hours per week: 30								
Total Credits: 23								

N.B. AC241307 is a Mandatory Audit Course (No Credit). It will be evaluated as PP (Pass) or NP (Not Pass)

DETAILED SYLLABUS

Course Code	Course Title	Hours per week L-T-P	Credit C
CE241301	Fluid Mechanics	3-0-0	3

Course Outcome (CO): After completion of the course, the students will be able to

- CO1 Identify the properties of fluids and various fluid pressure measurement techniques.
- CO2 Apply conservation laws to derive basic equations of fluid flow.
- CO3 Calculate the flow through openings and over hydraulic structures.
- CO4 Estimate different head losses in pipe flow system.
- CO5 Apply dimensional analysis to predict relationship among physical parameters that influence the flow in fluid mechanics.

MODULE 1: Fluid Properties

(3 Hours)

Fluid- definition, Important physical properties of fluid- mass density, specific weight, specific volume, specific gravity, viscosity- Newton's law of viscosity, surface tension, capillarity, Compressibility of fluids, Classification of fluids.

MODULE 2: Fluid Statics

(8 Hours)

Hydrostatic pressure, Pascal's law, Pressure height relationship, Absolute, gauge, atmospheric and vacuum pressure, Measurement of pressure- manometer, Hydrostatic forces on submerged plane and curved surfaces, Centre of pressure, Buoyancy, Metacentre and metacentric height, Equilibrium of floating bodies.

MODULE 3: Fluid Kinematics

(4 Hours)

Classification of fluid flow- steady and unsteady flow, uniform and non uniform flow, laminar and turbulent flow, compressible and incompressible flow, rotational and irrotational flow, one, two & three dimensional flow, Concept of stream lines, streak lines, path lines and stream tube, Stream function, Velocity potential function, Flow net, Free and forced vortex flow.

MODULE 4: Fluid Dynamics

(5 Hours)

Energy possessed by a fluid body in motion, Basic equations- continuity equation, energy equation (Euler's equation, Bernoulli's equation) and momentum equation, Application of energy equation and continuity equation- Venturimeter, orifice meter, pitot tube.

MODULE 5: Orifices and Mouthpieces

(4 Hours)

Classification of orifice, Flow through orifice, Hydraulic coefficients- definition and experimental determination, Discharge through large rectangular orifice, Discharge through fully submerged orifice, Discharge through partially submerged orifice, Time of emptying a tank (for both rectangular and hemispherical) through an orifice at its bottom; Classification of mouthpiece, Flow through an external cylindrical mouthpiece, Flow through a convergent- divergent mouthpiece.

MODULE 6: Notches and Weirs

(4 Hours)

Classification of notches and weirs, Discharge over a rectangular notch or weir, Discharge over a triangular notch or weir, Discharge over a trapezoidal notch or weir, Suppressed weir, Francis's formula with end contraction, Cippoletti weir, Discharge over a broad crested weir, Discharge over a narrow crested weir, Discharge over a submerged weir.

MODULE 7: Flow through Pipes

(4 Hours)

Loss of head due to friction, Darcy Weisbach formula, Minor head losses, Hydraulic gradient and total energy line, Flow through compound pipes, Equivalent pipe, Flow through parallel pipes, Siphon.

MODULE 8: Dimensional Analysis and Model Laws

(4 Hours)

Fundamental and derived quantities, Dimensional homogeneity, Methods of dimensional analysis- Rayleigh's method and Buckingham's pi-theorem, Model and prototype, Important dimensionless parameters and their significance, Geometric, kinematic and dynamic similarities, Distorted model and undistorted models.

Textbooks/References:

1. Modi, P. N. and Seth, S. M., Hydraulics and Fluid Mechanics (Including Hydraulic Machines), Standard Book House
2. Ramamrutham, S., Hydraulics, Fluid Mechanics and Fluid Machines, Dhanpat Rai Publishing Company
3. Bansal, R. K. A Textbook of Fluid Mechanics and Hydraulic Machines, Laxmi Publications
4. Ojha, C. S. P., Berndtsson, R., Chandramouli, P. N. Fluid Mechanics and Machinery Oxford University Press
5. White, F. M. Fluid Mechanics, McGraw Hill Education India Private Limited
6. Das, M. M. Fluid Mechanics and Turbomachines, PHI Learning
7. Garde, R. J. Fluid Mechanics through Problems New Age International
8. Subramanya, K., Theory and Application of Fluid Mechanics, Tata-McGraw Hill Publishing Co, New Delhi

Course Code	Course Title	Hours per week L-T-P	Credit C
CE241302	Engineering Survey	3-0-0	3

Course Outcomes:

At the end of the course, the students will be able to

- **CO1** Compare different methods of linear and angular measurements, adopt the most suitable one as per situation and detect and solve discrepancies in field data. Also will be able to evaluate the accidental errors involved in surveying using the concept of laws of accidental error and laws of weights.
- **CO2** Calculate, the Reduced Level (R.L.) of various points in the ground surface using different methods of contouring and draw contour lines & the areas and volume of various surveyed features.
- **CO3** Plan the setting out of a horizontal and vertical curve.
- **CO4** Plan a geodetic survey, considering strength of figure, intervisibility between stations
- **CO5** Use different instruments for different purposes using the knowledge of modern instruments as well as conventional equipment.
- **CO6** Take part in determination of scale, relief displacement and some ground information using the knowledge of aerial Photogrammetry and distinguish different ground features by the method of visual image interpretation using the knowledge of remote sensing.

MODULE 1: Introduction to linear measurements & Angular Measurements (6 Hours)

Introduction: Classification of survey, General principles.

Linear measurements: Instruments for linear measurements, ranging out survey line

Angular measurements: Bearing & meridian-types, Angular measurement using compass, local attraction, Magnetic declination, Angle measurement using theodolite, Traversing with compass and theodolite, Adjustments of closing errors

MODULE 2: Levelling & Contouring (6 Hours)

Levelling: Definition of basic terms, observation and reduction of Levels. Numerical problems. Profile levelling & Cross-sectioning, Fly Leveling, Curvature and refraction correction. Trigonometrical levelling Contouring: Definition, Characteristics of contours, Direct and indirect methods of contouring, Use of contour maps.

MODULE 3: Computation of area and volume (2 Hours)

Computation of area bounded by irregular boundaries- Mid ordinate rule, average ordinate rule, trapezoidal rule, Simpson's rule, Calculation of volume.

MODULE 4: Curve surveying (6 Hours)

Horizontal Curves: Different types of curves- Simple circular curve, Compound curve, Reverse curve, Transition curve, Length of transition curve, Ideal transition curve, Characteristics of transition curve, Setting out of simple circular and transition curve.

Vertical curves: Types, length, setting out of vertical curves.

MODULE 5: Triangulation (4 Hours)

Triangulation: Classification, Triangulation figures, Reconnaissance-selection of stations, Indivisibility of stations, Satellite station.

MODULE 6: Theory of errors and adjustments (4 Hours)

Errors and precision, definitions, Laws of accidental errors, Principle of least square, Laws of weights, Determination of most probable values- normal equations, method of correlates.

MODULE 7: Modern Survey Equipment (4 Hours)

EDM instruments: Electromagnetic wave theory, electromagnetic distance measuring system -

principle of working. Total station, Parts and accessories of total station, Field procedure for total stations.

GPS Surveying: Components, space segment, control segment and user segment, reference systems, GPS surveying techniques. DGPS, Principles of DGPS

MODULE 8: Photogrammetry and Remote sensing (8 Hours)

Photogrammetry: Introduction, Types of aerial photograph, Definitions, Scale of a vertical photograph, computation of length and height from vertical photograph, Relief displacement, Scale of tilted photograph, Flight planning, Stereoscopic vision, Parallax.

Remote sensing: Introduction, Electromagnetic Spectrum, interaction of electromagnetic radiation with the atmosphere and earth surface, remote sensing data acquisition: platforms and sensors.

TextBooks:

1. B.C. Punmia, Ashok K. Jain, Arun K Jain, Surveying Vol I, Surveying Volume II, Higher Surveying, Volume III, Laxmi publications, 2016.
2. S.K. Duggal Surveying Vol I, II, McGraw-Hill Education Pvt. Ltd., 2013.
3. R. Subramanian, Surveying and Levelling, Oxford University Press, 2015.
4. A M Chandra, Higher Surveying, New Age International Publishers.
5. S. K. Roy, Fundamental of Surveying, Prentice Hall India Learning.

Course Code	Course Title	Hours per week L-T-P	Credit C
CE241303	Construction Materials and Concrete Technology	3-0-0	3

Course Outcomes: At the end of this course, the students will be able to

- **CO1** Choose different types of construction materials (traditional and modern materials) for different purposes by studying different properties of the materials
- **CO2** Analyse different properties of fresh concrete and hardened concrete like workability, compressive strength, flexural strength, stress-strain behaviour, creep, durability etc. with reference to IS 456-2000
- **CO3** Take part in different testing of construction materials and concrete including NDT and prioritize the use of different NDTs according to the health of the structure
- **CO4** Design a concrete mix using the IS code method

MODULE 1: Traditional Materials: (4 Hours)

Historical development of construction materials, Traditional materials – Cement, Natural aggregates, Structural Steel and other metals, Bamboo, Timber and timber-based products, Bricks, Paints and Varnishes, Sealants and Adhesives; Glass and plastics, Ceramics and Refractories, Bitumen and asphaltic materials

Properties and Functional Requirements, BIS codes and specifications

MODULE 2: Innovative and Modern Materials (8 Hours)

M-Sand, Concrete (plain, reinforced and steel/glass fiber-reinforced, light-weight concrete, High Performance Concrete, Polymer Concrete, Self-healing concrete, Bendable concrete), Admixtures in Concrete Acoustic materials, Geo-textiles, laminates and adhesives, Graphene, Richlite, Liquid Granite, Carbon composites, Damp Proofing, Water Proofing, Thermal and Sound Insulation, Fire Protection materials, Solid and Hollow Concrete Block, Autoclaved Aerated concrete (AAC) Blocks, Fly-ash Bricks, Inter-locking Paver Blocks, Aluminum Composite Panels, Roofing (Corrugated Galvanized, Galvalume sheets).

Properties and Functional Requirements, BIS codes and specifications wherever applicable

MODULE 3: Cement, Aggregates, Steel (8 Hours)

Cement- Types and Grades of Cement, Hydration of cement, Physical and chemical properties, Gel Structure, Testing of Cement for Physical and chemical properties as per BIS specifications.

Aggregates-Testing of Aggregates for physical and mechanical properties, Water, Steel-Reinforced Steel Bars, Structural Steel.

MODULE 4: Fresh and hardened Concrete (10 Hours)

Proportioning of concrete, Concrete production, Workability and its measurement. Segregation, bleeding and laitance.

Strength of Concrete: Compressive strength and factors affecting it, Testing of hardened concrete-cube and cylindrical sample, Platen effect, flexure test, Stress-strain relation and modulus of elasticity, Shrinkage, Creep of concrete and its effect.

Durability of Concrete: Corrosion of reinforcing bars, sulphate attack, frost action, deterioration by fire, concrete in seawater, acid attack, and carbonation

MODULE 5: Concrete Mix Design (12 Hours)

Principles and methods, Statistical Quality control. Concrete Rheology, Maturity concept, IS method for concrete Mix Design

MODULE 6: Non-destructive testing (3 Hours)

Destructive vs. Non-Destructive testing, Methods & Principles of NDT. Rebound hammer, UPV, core-cutting.

Textbooks/References:

1. S. C. Rangawalla, Engineering Materials, Charotar Publishing House Pvt. Ltd. 2017.
2. P. C. Varghese, Building Materials, PHI Learning 2015.
3. K.S. Nanjunda, K S Jagadish, B.V. Reddy, Rao Venkatarama, Alternative Building Materials and Technologies New Age International Private Limited, 2023.
4. Rai Mohan and Jai Singh M.P., Advances in Building Materials and Construction, CBRI Roorkee, 1986.
5. Technical Teachers Training Institute Chandigarh, Civil Engineering Materials, Tata McGraw Hill.
6. Robin J. S. Spence, David J. Cook, Building Materials in Developing Countries, John Wiley and Sons, 1983.
7. M. S. Shetty Concrete Technology, Theory and Practices, S.Chand and Company Ltd., 2019.
8. A. M. Neville, Properties of Concrete Pearson Education India, 2012.
9. M. L. Gambhir, Concrete Technology: Theory And Practice, McGraw Hill Education, 2017.
10. M. L. Gambhir, Concrete Manual , Dhanpat Rai & Co, 2012.

Course Code	Course Title	Hours per week L-T-P	Credit C
CE241304	Solid Mechanics	3-0-0	3

Course Outcomes: At the end of the course, the students will be able to:

- **CO1** Solve Simple Stress and Strain problems, along with Mohr's circle for a given plane stress/strain condition.
- **CO2** Solve problems on Bending Stress and Shear Stress in beam
- **CO3** Solve for Axial and Bending stresses in a column
- **CO4** Solve problems on Thin Cylinders and Shells
- **CO5** Solve problems on Torsion in solid and hollow circular shafts

MODULE 1: Simple stress and strain

(10 Hours)

Simple Stress and Strain: Tensile, Compressive and Shear Stress, Stress-strain diagrams for brittle and ductile materials, Hooke's law, Young's modulus, Poisson's ratio, Elastic Constants and their relationship, Principal Stresses and Strains; Mohr's Circle for Plane-Stress and Plane-Strain, Generalized Stress and Strain.

MODULE 2: Theory of Simple Bending

(10 Hours)

Theory of Simple Bending - Bending stresses in beams, Shear stress distribution in various cross sections of beam.

MODULE 3:

(10 Hours)

Introduction to the concept of Elastic Instability, Slenderness ratio, Critical load of Columns: Concentric and eccentric, Columns with initial curvature, Equivalent eccentricity, Empirical formula for critical load of column, Rankine's formula. Beam-Column.

MODULE 4:

(7 Hours)

Membrane stress in Thin Cylinders and Shells under internal fluid pressure.

MODULE 5:

(8 Hours)

Theory of Pure Torsion in Solid and Hollow circular shafts - Torsional Shear Stresses.

Text Books:

1. S. P. Timoshenko/D. H. Young, Elements of Strength of Materials, EWP, 2003.
2. Irving H. Shames/James M. Pitarresi, Solid Mechanics, Pearson Education India: 2015.
3. R. Subramanian, Strength of Materials, Oxford University Press, 2016.
4. S.M.A. Kazimi, Solid Mechanics, McGraw Hill Education, 2017.
5. NPTEL(MOOCs): NPTEL study material for Solid mechanics.

Course Code	Course Title	Hours per week L-T-P	Credit C
MA241305A	Mathematics III-A	3-0-0	3

Course Outcomes:

At the end of this course, the students will be able to

- **CO1** Solve first- and second-order PDEs using analytical methods and classify second-order equations.
- **CO2** Apply separation of variables to solve diffusion, wave, and Laplace equations with given conditions.
- **CO3** Understand and apply the concept of numerical methods in solving real life problems.
- **CO4** Apply probability theory, including discrete and continuous distributions, expectation, moments, and Bayes' rule, to model uncertainty and analyze random phenomena in engineering contexts.
- **CO5** Apply Laplace and inverse Laplace transforms and Fourier transforms, including their properties and convolution theorem, to solve ordinary differential equations arising in engineering applications.
- **CO6** Apply statistical techniques—including data analysis, curve fitting, correlation, and hypothesis testing—to interpret and make decisions from experimental and observational data.

MODULE 1: Partial Differential Equations

(10 Hours)

Definition of Partial Differential Equations, First order partial differential equations, solutions of first order linear PDEs; Solution to homogenous and non-homogenous linear partial differential equations of second order by complimentary function and particular integral method. Second-order linear equations and their classification, Initial and boundary conditions, Separation of variables method to simple problems in Cartesian coordinates. One dimensional diffusion equation, wave equation, Laplace equation and their solution by separation of variable method.

MODULE 2: Numerical Methods

(10 Hours)

Gauss-Seidal iteration method to solve a system of equations - Numerical solution of algebraic and transcendental equations by Regula-Falsi method and Newton-Raphson's method - Lagrange interpolation, Forward and backward differences, Newton's forward and backward interpolation formulae - Numerical differentiation with forward and backward differences - Numerical Integration with Trapezoidal rule, Simpson's 1/3 rule and Simpson's 3/8 rule - Taylor series method, Euler's method, 4th order Runge-Kutta method for solving first order ordinary differential equations.

MODULE 3: Probability

(10 Hours)

Probability spaces, conditional probability, independence; Discrete random variables, Independent random variables, the multinomial distribution, Poisson approximation to the binomial distribution, infinite sequences of Bernoulli trials, Expectation, Moments, Variance, Chebyshev's Inequality. Continuous random variables, distribution functions: normal, exponential and gamma densities. Bivariate distributions, Bayes' rule.

MODULE 4(a): Laplace Transform and Fourier Transform

(10 Hours)

Laplace transform of elementary function, Properties of Laplace transform, Inverse Laplace transform (without proof of formulae), Convolution theorem (without proof of the theorem), Solution of ordinary differential equations with the help of Laplace transform.

Fourier Transform:

Solution of partial differential equation by Fourier Transform

MODULE 4(b): Statistics

Basic Statistics, Measures of Central tendency: Moments, skewness and Kurtosis Correlation and regression – Rank correlation. Curve fitting by the method of least squares- fitting of straight lines, second degree parabolas and more general curves. Test of significance: Large sample test for single proportion, difference of proportions, Tests for single mean, difference of means, and difference of standard deviations. Test for ratio of variances - Chi- square test for goodness of fit and independence of attributes

Remarks:

- 1. Module 1, 2, 3, 4a: for Chemical Engineering**
- 2. Module 1, 2, 3, 4b: for Civil Engineering, Mechanical Engineering, Industrial and Production Engineering**

Text Books:

1. Reena Garg, Engineering Mathematics, Khanna Book Publishing Company, 2022.
2. Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021.
3. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, 9th Edition, 2006.
4. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint 2010.
5. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003.
6. S. Ross, A First Course in Probability, Pearson Education India , 6th Ed., 2002

Course Code	Course Title	Hours per week L-T-P	Credit C
HS241306	Indian Knowledge Systems	2-0-0	2

Course Objectives:

1. To introduce students to the foundational concepts of Indian Knowledge Systems (IKS).
2. To explore the contributions of ancient and medieval India in science, technology, mathematics, and philosophy.
3. To develop an appreciation for traditional Indian knowledge and its relevance in modern contexts.
4. To encourage critical thinking about indigenous knowledge systems and their applications.

Course Outcome (CO):

After completion of the course, the students will be able to

- **CO1:** Understand the key principles of Indian Knowledge Systems.
- **CO2:** Recognize India's contributions to science, mathematics, medicine, and engineering.
- **CO3:** Analyze the interdisciplinary nature of IKS and its connections with modern technology.
- **CO4:** Appreciate the philosophical and ethical dimensions of traditional Indian knowledge.

MODULE 1: Introduction to Indian Knowledge System (IKS) (4 Lectures)

- Definition, scope, and significance of IKS
- Sources of Indian knowledge: Vedas, Upanishads, Puranas, Smritis, and other texts
- Classification of knowledge in Indian tradition (Vidyas and Kalas)

MODULE 2: Science & Technology in Ancient India (12 Lectures)

- Contributions to mathematics (Zero, Decimal System, Vedic Mathematics, Aryabhata, Brahmagupta)
- Astronomy & Space Science (Surya Siddhanta, Planetary Motion, Concept of Time)
- Metallurgy and material sciences (Iron Pillar, Wootz Steel, Ayurvedic alloys)
- Civil engineering & architecture (Temple architecture, Town planning, Water management)

MODULE 3: Indian Philosophical & Ethical Systems (6 Lectures)

- Key schools of Indian philosophy (Nyaya, Vaisheshika, Samkhya, Yoga, Vedanta)
- Concept of Dharma, Karma, and sustainable living
- Ethics in science and technology from an Indian perspective

MODULE 4: Traditional Medicine & Wellness Systems (4 Lectures)

- Foundations of Ayurveda, Yoga, and Siddha
- Holistic health and wellness practices
- Relevance of traditional medicine in modern healthcare

MODULE 5: IKS in Contemporary Applications (4 Lectures)

- Revival of traditional knowledge in modern science and engineering
- Case studies (e.g., Panchakarma in wellness tech, Vastu in architecture)
- Government initiatives (National Education Policy 2020, IKS Division under AICTE)

Text books:

1. "Indian Knowledge Systems" – Kapil Kapoor & Avadhesh Kumar Singh
2. "Ancient Indian Leaps into Mathematics" – B.S. Yadav
3. "Science and Technology in Ancient India" – O.P. Jaggi
4. "The Positive Sciences of the Ancient Hindus" – Brajendranath Seal
5. AICTE Model Curriculum for IKS (Available online)

Online Resources:

1. NPTEL / SWAYAM courses on IKS
2. AICTE-IKS Portal (<https://iksindia.org/>)
3. Digital Library of India (<https://dli.gov.in>)

Course Code	Course Title	Hours per week L-T-P	Credit C
AC241307	Language other than MIL	1-1-0	0

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Demonstrate a clear understanding of basic grammar rules including subject-verb agreement, tenses, and cases, and apply them in constructing grammatically correct sentences.
- **CO2:** Identify and correct common grammatical and usage errors in written English, enhancing sentence effectiveness and clarity.
- **CO3:** Enhance their vocabulary through the use of synonyms, antonyms, idiomatic expressions, and phrasal verbs, improving both written and spoken communication.
- **CO4:** Apply verbal reasoning skills to comprehend, arrange, and complete sentences and paragraphs effectively.
- **CO5:** Develop strong reading comprehension skills using skimming, scanning, and intensive reading techniques to extract key information from various texts.
- **CO6:** Produce coherent and concise writing through precis writing and essay composition, demonstrating logical flow, clarity, and appropriate structure.

MODULE 1: Basic grammar usage

(10 Lectures)

Subject-verb agreement, narration, tenses

Case: Subjective case, possessive case, objective case

Writing effective sentences, Spotting errors

MODULE 2: Vocabulary Building

(6 Lectures)

Synonyms and Antonyms, One word substitution,

Idioms and Phrases and Phrasal Verbs

MODULE 3: Verbal ability

(8 Lectures)

Cloze test

Sentence rearrangement, Sentence completion,

Paragraph Jumbles

MODULE 4: Writing and Reading Ability

(8 Lectures)

Reading subskills, Skimming, Scanning and Intensive reading

Reading Comprehension, Precis Writing, Essay Writing

Text books

1. Objective General English: S.P.Bakshi, Arihant Publication India Ltd., 2021
2. Descriptive English: S.P.Bakshi, Arihant Publication India Ltd., 2016

Course Code	Course Title	Hours per week L-T-P	Credit C
CE241317	BCP Laboratory	0-1-2	2

Course Outcomes: Students will be able to:

- **CO1** Understand various components of a building such as Foundation, Sub-structure and Superstructure
- **CO2** Formulate the sequence of construction to be implemented at real site
- **CO3** Identify and use of various the construction tools, equipment machineries required in construction
- **CO4** Do the layout of a building on field based on layout plan using construction drawings
- **CO5** Do planning of a building using Bye-laws issued by local authorities

MODULE 1: Building Components (6 Hours)

Introduction to building construction with Demonstration of various building components such as shallow and deep foundation, Plinth, Columns, Beams, roof, roof coverings

MODULE 2: Building Drawings (6 Hours)

Study of various building drawings such as, Layout plan, Architectural Drawings, Structural drawings, utility drawings.

MODULE 3: Construction tools and equipment (4 Hours)

Demonstration of various construction tools, equipment and machineries used in building construction.

MODULE 4: Layout (6 Hours)

On field layout of a building according to the layout plan provided

MODULE 5: Building Planning (8 Hours)

Orientation of Building, Concept of Green Building, Use of Vastu in Planning. Planning of a RCC building following the latest local by-law.

Textbooks/References:

1. B.C. Punmia, Building Construction, Laxmi Publications (P) Ltd.
2. GMDA Building Byelaw.
3. GMDA Master Plan.
4. National Building Code of India (2005), BIS.
5. P.C. Vargesse , Building Construction, Practice Hall of India, New Delhi.
6. Malik & Meo Civil Engineering Drawing Computech Publication ltd.
7. Gurcharan Singh / Jagdish Singh Building Planning, Designing & Scheduling Standard Publications.
8. Jagadish, Reddy and Rao, Alternative Building Materials and Technologies, New Age International (P) ltd.

Course Code	Course Title	Hours per week L-T-P	Credit C
CE241313	Construction Materials and Concrete Technology Lab	0-0-2	1

Course Outcomes:

After completing this course, the students will be able

- **CO1** Evaluate the physical properties of cement
- **CO2** Determine different properties of coarse and fine aggregates
- **CO3** Examine the properties of fresh and hardened concrete like workability and compressive strength.

LIST OF EXPERIMENTS

MODULE 1: Testing of cement

(4 Hours)

Standard consistency,
Setting time (initial and final),
Fineness and soundness
Compressive strength test (3 days, 7 days and 28 days).

MODULE 2: Testing of Fine Aggregates

(4 Hours)

Sieve analysis for zoning and fineness modulus (FM)
Bulking of sand
Absorption and moisture content
Specific gravity

MODULE 3: Testing of Coarse aggregate

(4 Hours)

Sieve analysis for grading
Absorption and moisture content
Specific gravity
Flakiness index, Elongation index

MODULE 4: Tests on fresh concrete

(4 Hours)

Compressive strength test of concrete
Workability test of fresh concrete

MODULE 5: Concrete Mix Design by IS method and subsequent testing of cubes

(4 Hours)

Textbooks/References:

1. Laboratory Manuals
2. E-content

Course Code	Course Title	Hours per week L-T-P	Credit C
CE241312	Engineering Survey Lab	0-0-2	1

Course Outcome:

After completing this course, the students will

- **CO1** Measure the linear distance between two points.
- **CO2** Carry out closed compass, theodolite (Gale's Traverse Table) ,plane table, GPS and Total Station traversing of an area and draw a map.
- **CO3** Draw Cross Section, Profile, Contour Map.
- **CO4** Determine height of an object and vertical/horizontal angles by operating a transit theodolite
- **CO5** Set out a Simple circular curve using method of offset & Rankine's Method.
- **CO6** Set out a Combined curve (Simple circular and Transition curve).
- **CO7** Describe the process of DGPS, Drone Survey and use of associated software.

LIST OF EXPERIMENTS

1. To practice Chaining and tape methods of ranging and to measure the distances between two points (Direct ranging).
2. To measure bearings of sides of traverse with prismatic compass and computation of included angles.
3. To carry out the Profile levelling and cross-sectioning.
4. To prepare the contour map of an area by indirect method.
5. Horizontal angle measurement by Theodolite and determination of the height of an object with Trigonometric levelling.
6. Curve setting: Simple circular curve by Rankine's method.
7. Curve setting: Combined curve: To set a combined curve (Simple circular and Transition curve) between two given straight roads.
8. To carry out open traversing with Total Station.
9. Demonstration of DGPS Surveying and Associated Software
